

NETWORKS AND COMPLEXITY

Solution 17-2

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 17.2: Codimension example [Lvl. 2]

Consider the system

$$\dot{x} = f(x, a, b) = a - bx^2.$$

For $b > 0$ find the steady states and their bifurcation points. Plot the bifurcation points in a 2-parameter bifurcation diagram, where the axes are the parameters a and b .

Solution

We compute the steady states

$$0 = a - bx^2 \quad (1)$$

$$bx^2 = a \quad (2)$$

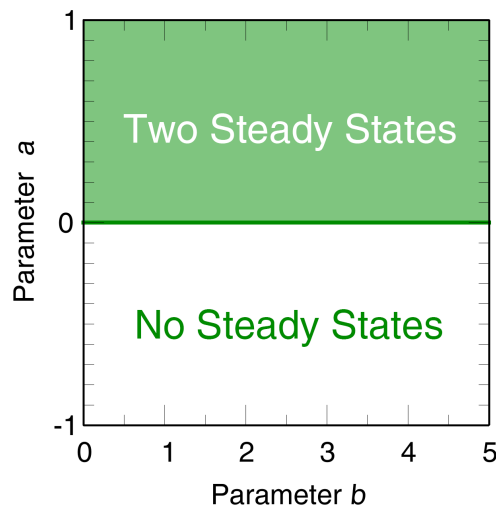
$$x^2 = a/b \quad (3)$$

$$x = \pm\sqrt{a/b} \quad (4)$$

So for $b > 0$ and $a > 0$ we have two branches of steady states $x_{1,2} = \pm\sqrt{a/b}$. At $a = 0$ these two branches collide in a square root shape and annihilate each other so that there are no steady states left. This already shows that there is a fold bifurcation at $a = 0$ regardless of b , as long as $b > 0$. If we want to do another test we could compute

$$f_x = \frac{\partial \dot{x}}{\partial x} = -2xb = \pm 2\sqrt{ab} \quad (5)$$

which becomes zero at $a = 0$, confirming the presence of the bifurcation point. We can thus plot the two-parameter diagram:



As a bonus it is interesting to explore what happens at $b = 0$. The line $b = 0$ is also a bifurcation line as $f_x = 0$. However it is a very weird bifurcation as our steady states disappear to infinity. For $b < 0$ the steady states reappear, but they now exist for $a < 0$. So this bifurcation line is none of the bifurcations that we studied so far but a very highly degenerate case, where almost all derivatives of f become zero. such highly degenerate bifurcations can appear in very simple examples, because the most simple systems tend to be very special, and hence degenerate.

The point $a = b = 0$ where the degenerate bifurcation intersects the fold is an example of a (highly degenerate) codimension-2 bifurcation where the fold bifurcation changes its nature from creating two steady states on the positive- a side, to creating two steady states on the negative- a side of the bifurcation manifold.